

Al Neelain University

Faculty of Engineering

Department of Computer Engineering

Presented by student:

Set Al-Nisa Ismail Mohammed Abdel Zahir

Supervised by:

Dr. Ahmed Al-Fatih



C PROGRAMMING LANGUAGE

Store Management System

Store Management System

A practical project for managing store products — add, view, search, and delete — using fundamental data structures in the C language.

Prepared by: ست النساء إسماعيل محمد عبد الظاهر



Project Overview



Objective

Build a Console Application to manage product inventory in the store efficiently and easily.

Core Features

- Add new products with full details (ID, name, quantity, price)
- Display a neatly organized list of all registered products
- Search for a product using its ID
- Delete a product from the system

Used Technologies

- `struct` — for storing product data
- `Arrays` — for temporary storage
- `Functions` — for organizing the code

Data Structure — Data Structure

struct was used to organize a single product's data, combining different data types into one coherent entity.

```
struct Product {  
    int id;    // Product ID  
    char name[50]; // Product name  
    int quantity; // Available quantity  
    float price; // Price  
};
```

Global Variables

Products Array

```
struct Product  
products[MAX_PRODUC  
TS];  
Supports up to 100  
products
```

Product Counter

```
int count = 0;  
Tracks the current  
number of products in  
the system
```

Add and Display Products

+ Add Product — `addProduct()`

- First checks whether the inventory is full (`count >= MAX_PRODUCTS`)
- Requests input step by step: ID, name, quantity, price
- Uses `fgets` to read names safely and avoid whitespace issues
- Increases `count` by 1 on success

📋 Display Products — `displayProducts()`

- Checks whether the store is empty (`count == 0`)
- Loops through all products with a `for` loop
- Displays the details in a clean, easy-to-read table format



Search and Delete

Search for a Product — `searchProduct()`

- Prompts the user to enter the product ID
- Searches using an iterative loop to match the ID
- Displays the details as soon as the product is found
- Prints "**Product not found**" when no match is found

Delete a Product — `deleteProduct()`

- Searches for the product by the required ID
- Shifts the following elements to fill the gap:
`products[j] = products[j + 1]`
- Decreases `count` by 1 to update the total

 Shifting elements is necessary to keep the array sequence without gaps.

Managing Input and the Buffer

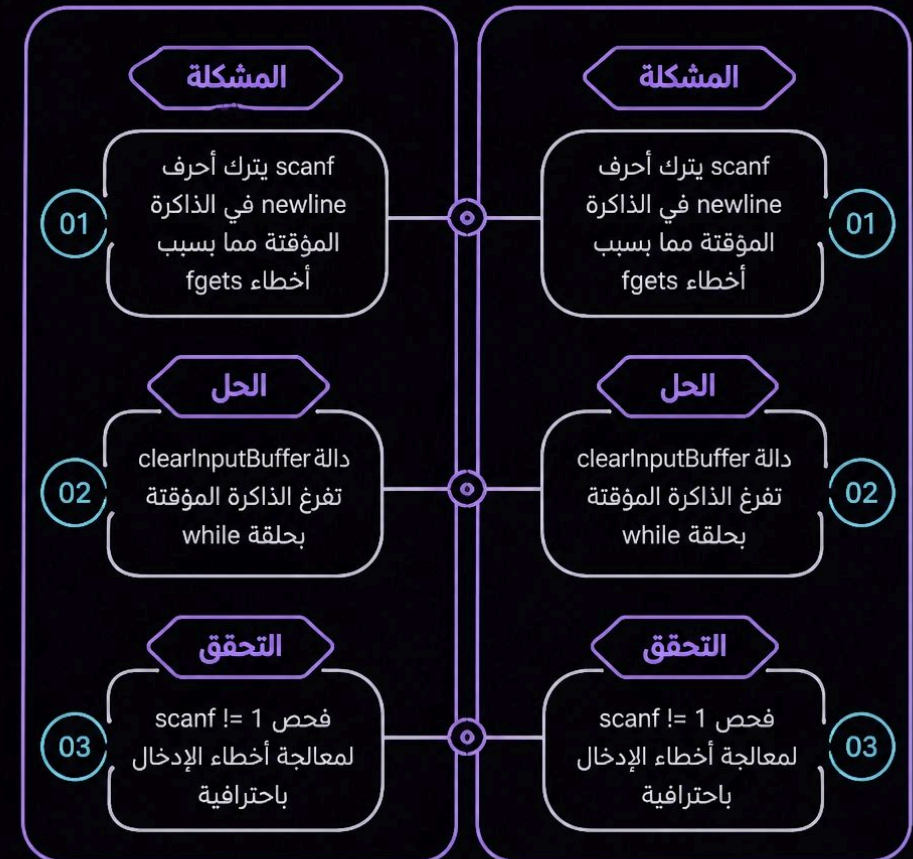
The Problem

When using `scanf` and then moving to `fgets`, unwanted characters (such as `\n`) remain in the buffer, which causes input errors.

The Programmatic Solution

A helper function `clearInputBuffer()` was created to clear the buffer before each new read.

```
void clearInputBuffer(void) {  
    int c;  
    while ((c = getchar()) != '\n'  
           && c != EOF) {}  
}
```



Main Function — Main Loop

It relies on an infinite `while(1)` loop that continuously displays the menu options until the user chooses to exit.

1

Add Product

Add a new product

2

Display Products

Display all products

3

Search Product

Search by product ID

4

Delete Product

Delete a product from the system

5

Exit

End the program

The `switch-case` statement is used to guide the user and execute the appropriate function based on their choice.

Conclusion and Future Enhancements

✓ What Was Achieved

- Comprehensive practical application of C language concepts: **Structures, Arrays, Control Flow**
- Good error handling and professional input management
- Well-organized code divided into clear, responsible functions

🚀 Suggested Future Additions

Data Persistence

Connect the program to files or a database to save data permanently after closing

Edit Products

Add an Update Product function to modify the data of an existing product

Graphical Interface

Move from a Console Application to a graphical interface using libraries such as ncurses or GTK

Store Management System Workflow

The following diagram shows the continuous flow of the main functions in the store management system.



The cycle begins with adding products and continues by providing effective tools for inventory management.